

# GOURI ANIL

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## EDUCATION

LOUISIANA STATE UNIVERSITY

**PhD Candidate**, Department of Oceanography and Coastal Sciences

CGPA: 4.18

Baton Rouge, LA

August 2022 - August 2026

Expected

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

**MS in Physical Oceanography**

CGPA: 9.69

Kerala, India

August 2019 - June 2021

MAR ATHANASIOS COLLEGE OF ARTS AND SCIENCES

**BS in Physics**

CGPA: 9.43

Kerala, India

June 2016 - May 2019

## RESEARCH EXPERIENCE

LOUISIANA STATE UNIVERSITY

PhD Candidate

Baton Rouge, LA

August 2022 - Present

***Assessing Impacts of Stratospheric Aerosol Injection (SAI) on Marine Thermal Stress for Ecosystems***

Advisor: Dr. Cheryl S Harrison

Committee: Dr. Daniel Holstein, Dr. Junhong Liang, Dr. Gaurav Kandlikar

*Project 1: Impacts of Timing and Level of SAI on Coral Thermal Bleaching*

*Project 2: Marine Climate Exposure under SAI and Assessing Socio-economic Impacts to Global Nations*

*Project 3: Building an Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of SAI*

INDIAN NATIONAL CENTRE FOR OCEAN INFORMATION SYSTEMS (INCOIS)

Graduate Researcher

Advisor: Dr. Abhisek Chatterjee (Senior Scientist, INCOIS)

Hyderabad, India

December 2020 - May 2021

*Understanding Marine Heatwaves in the Arabian Sea*

INDIAN INSTITUTE OF TECHNOLOGY

Undergraduate Summer Research Fellow

Advisor: Dr. Varun Bhalerao

Mumbai, India

May 2018 - July 2018

*X-Ray data analysis of a pulsar Centaurus X-3 using NASA HEASoft*

## EXTRAMURAL FUNDING

2025 **Environmental Defense Fund**, Concept Lead and Lead Proposal Writer (Funded)

*Preventing Coral Bleaching Through SAI: An Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of Stratospheric Aerosol Injection*

PI: Dr. Cheryl S Harrison, Co-PI: Dr. Thomas Oliver

## PUBLICATIONS

### [Google Scholar](#)

#### **Published**

A. Chatterjee, **G. Anil**, L. R. Shenoy, *Marine Heatwaves in the Arabian Sea*, Ocean Science, 2022, DOI: 10.5194/os-18-639-2022

K. E., Roberts, T. Rohr, ..., **G. Anil**, ...C.S. Harrison (2026). Potential Impacts of Climate Interventions on Marine Ecosystems. Reviews of Geophysics, 64(1), e2024RG000876. <https://doi.org/10.1029/2024RG000876>

#### **Submitted**

**G. Anil**, C.S. Harrison, J.A. Kleypas, D. Holstein, T.A. Oliver, V. Garza, K.E. Roberts, T. Felgenhauer, D. Jayewardene, *Impacts of Timing and Level of Stratospheric Aerosol Injection on Coral Thermal Bleaching*, ESS Open Archive. December 17, 2025. DOI: 10.22541/essoar.176599087.79155141/v1

L. Kounta, L. Luo, **G. Anil**, D. Hueholt, C. S. Harrison, D. Visioni, M. Tye, T. Felgenhauer, A. T. Gaye, P.L. Zarnetske, *Climate Intervention through Stratospheric Aerosol Injection may partially mitigate marine heatwaves*, ESS Open Archive. November 29, 2025. DOI: 10.22541/essoar.176442822.27571009/v1

## PRESENTATIONS

### **ORAL**

|      |                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | AGU Fall Meeting<br>LSU 3-Minute Thesis Competition Finals<br>LSU College of Coast and Environment Annual Graduate Student Symposium |
| 2024 | Reef Futures Symposium                                                                                                               |

### **POSTERS**

|      |                                  |
|------|----------------------------------|
| 2025 | AGU Fall Meeting                 |
| 2024 | LSU Graduate Research Conference |
| 2024 | AGU Ocean Sciences Meeting       |

## MEDIA COVERAGE ON GRADUATE RESEARCH

|      |                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Could Stratospheric Aerosol Injection Help Save Corals from Bleaching? Eos. <a href="#">Article</a>                                             |
| 2023 | Marine heat waves may be favouring more cyclones in the Arabian Sea: INCOIS scientists. <i>The Hindu</i> . <a href="#">Article</a>              |
| 2022 | 'At least 20 marine heatwave days in Arabian Sea during last decade.' <i>The Times of India</i> . <a href="#">Article</a>                       |
| 2022 | The Arabian Sea faces an increased frequency and duration of marine heatwave days, finds study. <i>Mongabay-India</i> . <a href="#">Article</a> |

## AWARDS AND FELLOWSHIPS

|      |                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------|
| 2025 | Travel Award to attend the 16 <sup>th</sup> Geoengineering Model Intercomparison Project Meeting, Tokyo |
| 2025 | NCAR High-Performance Computing Large Allocation for Climate Intervention Simulations                   |
| 2025 | Travel Award to attend the 2025 Degrees Global Forum, South Africa                                      |
| 2024 | Graduate Student Organization Travel Award                                                              |
| 2023 | Travel Award to attend the Scripps Marine Geoengineering Workshop                                       |
| 2022 | Graduate Excellence Award for Academic Achievement                                                      |
| 2019 | Indian Academy of Sciences Summer Research Fellowship                                                   |
| 2018 | Indian Academy of Sciences Summer Research Fellowship                                                   |

## PEER REVIEW

|      |                                                                              |
|------|------------------------------------------------------------------------------|
| 2025 | Co-reviewer for Journal of Geophysical Research – Oceans                     |
| 2025 | Reviewer for American Academy of Arts & Sciences Solar Geoengineering Primer |

## **TEACHING AND MENTORING**

- 2026 Teaching Assistant and Guest Lecturer, Gulf of Mexico (Spring Semester)
- 2025 Undergraduate Honors Thesis Mentor (December 2025 to April 2026)
- 2024 Teaching Assistant, Introduction to Oceanography (Fall Semester)
- 2023 Teaching Assistant, Introduction to Oceanography (Fall Semester)

## **WORKSHOPS**

- 2024 Community Climate Intervention Strategies (CCIS) Workshop, Notetaker
- 2023 Scripps Marine Geoengineering Scenario Design Workshop, Rapporteur
- 2019 International Conference on Frontiers in Marine Science, Volunteer and Session Host

## **OUTREACH**

- 2024 Student Host for Ocean Commotion Outreach Event by LSU
- 2024 Student Host for LSU College of Coast and Environment Open House
- 2024 Coast and Environment Graduate Organization Secretary
- 2018 Volunteer and Team Lead, Guardians of Dreams Non-profit, India
- 2016 Volunteer and HR Team Member, The Art Outreach Society Non-profit, India

## **TECHNICAL SKILLS**

Programming: Python, R

Modeling: Experience running Community Earth System Model Simulations, Model Data Processing

Climate Data Processing and Visualization: netCDF, xarray, geopandas, matplotlib, CDO

Spatial Tools: QGIS